

Hoang Minh Thanh

WebGL Front-End Engineer

hmthanh.github.io [hmthanh](#) [in](#) [hmthanh](#) [Vietnam](#) hmthanhgm@gmail.com

A front-end engineer with a strong foundation in real-time 3D graphics and performance optimization, proficient in WebGL, Three.js, and JavaScript. Experienced in developing interactive web-based visualizations and managing large-scale 3D datasets like 3D Tiles and glTF. I am well-suited to contribute to the cutting-edge 3D visualization tools at **Intratech Corp.**

★ Keypoint

- **Expert in WebGL, Three.js, Nextjs, React:** Proven experience developing and optimizing complex, interactive 3D web applications. Developed scalable projects like [MillionScope](#) and [MetaPet](#), focusing on efficient rendering and managing large 3D datasets. Deep understanding of the WebGL pipeline and shader programming ([GLSL](#)).
- **Proficient in 3D Data Pipeline & Asset Optimization:** Strong practical knowledge of 3D formats like [glTF](#) and workflows for the web. Skilled in Blender and Maya for modeling, optimization, and technical asset preparation, ensuring efficient loading and rendering of large-scale industrial models for browser-based visualization.
- **Strong Foundation in 3D Graphics Principles & Performance:** Comprehensive theoretical knowledge applied to practical problems. Studied advanced concepts from "Ray Tracing in One Weekend," [PBRT](#), and [BRDF implementation](#). This foundation is crucial for architecting performance-critical optimizations for memory management and rendering complex scenes in WebGL.

🎓 Education

AS	University of Science - VNUHCM, Information Technology	2012 – 2015
BS	University of Science - VNUHCM, Computer Science	Sept. 2018 – Sept. 2020
	• GPA: 3.3/4.0 • Thesis: GCAT - Link Prediction in Knowledge Graph, [Report], [Code]	
MSc	University of Science - VNUHCM, Computer Science	Oct. 2021 – Dec. 2024
	• GPA: 3.44/4.0 • Thesis: OpenHuman: A conversational gesture synthesis system based on emotions and semantics, [Report], [Code]	

📖 Publications

DeepGesture: A conversational gesture synthesis system based on emotions and semantics [arXiv] , [homepage]	Jul. 2025
Towards a GENE Leaderboard [arXiv]	Oct. 2024
Graph Collaborative Attention Network for Link Prediction in Knowledge Graphs [arXiv] [homepage]	Oct. 2020

💼 Experience

Rakumo Inc, Software Engineer	Jul. 2019 – Apr. 2020
• Developed a Django-based web application with integrated DocuSign.	
FPT Software, AWS Data Engineer	Oct. 2020 – Nov. 2021
• CI/CD pipeline terraform in Jenkins, implemented and modified real-time and	

batch data pipelines on AWS using Kinesis, Lambda, S3, Glue, StepFunction, SNS, ..etc..

- Cox Automotive (onsite-offshore): Collaborated with international teams in Agile

VNG Corp, Software Engineer

Oct. 2021 – May 2024

ZDN Team (Zalo Content Delivery Network): caching service serve millions request

- Implement scheduled `ScheduledThreadPool` for monitoring stats execution.
- Built Ant admin dashboard with server visualization via amchart, p5.js in NextJS.
- Upgrade internal Golang package for Apache Thrift

OpenHuman (openhuman.ai), Creator

May 2024 – Present

- Join GENE Research Team ([GENEA Leaderboard](#), [HEMVIP](#)): Designed and built a full-stack, production-ready leaderboard system to evaluate multiple AI gesture generation models, integrating user study workflows via Prolific.

</> Selected Projects

DeepGesture ([Code](#))

deepgesture.github.io

Three.js realisitic face for OpenHuman ([Code](#))

renderengine.pages.dev

- Loaded and scaled facial OBJ parts in Blender (eyeball, head, tongue, etc.); applied edited albedo, roughness, specular, clearcoat maps; configured dat.GUI, environment toggle, and tone mapping for realistic facial rendering

HairMimic - In progress hair simulation for OpenHuman ([Code](#))

hairmimic.github.io

- Generate hair from Maya XGen Grooming and load hair meshes.

MillionScope ([Code](#)) ChatGPT-Style streaming response

millionscope.com

Unity Game Pucca Runner ([Code](#)) Pathfinding with a 2D hash map and Dijkstra's algorithm; runner moves toward the player and triggers collision on boundary breach

[Pucca Runner](#)

Three.js Demo Project: [GlowShader/ Code](#), [SpringSphere/ Code](#), [Vinfast Car/Code](#), [Jumming Girl/ Code](#), [Cross Sphere/ Code](#).

Others projects: [Morph target animation ReadyPlayerMe](#), [MetaPet \(metapet.vercel.app\)](#), [cortexpod.com](#); [pi-thagon.com](#), More on: github.com/hmthanh

🌟 Certifications

- [Critical Thinking Certification 2014](#), [Emotion Recognition Certification 2019](#), [Summer ML](#)
- Coursera: [Design Pattern](#), [TensorFlow Developer Professional](#) 2021
- **AWS:** [Solutions Architect - Associate](#), [Machine Learning - Specialty](#); [Solution Architect](#) 2021

💻 Technologies

Core 3D Web: WebGL, Three.js, 3D Tiles, glTF/GLB, GLSL Shader

Programming & APIs: JavaScript, TypeScript, Python, C++

3D Tools & Optimization: Blender, Maya (for modeling & asset optimization), Unity

Key Expertise: Large-scale 3D Data Visualization, Real-time Rendering Optimization, Browser-based 3D Application Development